AI Image Generator

10. Change Log and Roadmap

Version 1.0
22 October 2026

Project owner: Frazile AI
Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Stable Diffusion XL + Real-ESRGAN, served locally with Gradio, with per-image GPU telemetry recorded in SQLite.

10. Change Log and Roadmap

10.1 Change log

Increment 1: command-line generator (21 Oct 2026)

- Created the project folder, virtual environment, and requirements.txt.
- Installed torch 2.14.1+cu126, diffusers 0.41, transformers, accelerate.
- Wrote generate.py with prompt, negative, steps, guidance, size, seed, count and model arguments.
- First image: 1024x1024 in 9.7 s, peak VRAM 10.5 GB.
- Fixed: VAE slicing call renamed in diffusers 0.41.

Increment 2: web interface

- Added app.py using Gradio 6.30 with prompt, negative prompt, style and size dropdowns, step and guidance sliders, seed and count, a gallery and an info box.
- Model loaded once at startup and kept resident.
- Added run-app.bat and README.

Increment 3: prompt-driven controls

- Removed style, size, steps and guidance controls.
- Added prompt_parser.py: ratios, exact sizes, orientation words, quality and fidelity words, twelve style presets, count, seed, steps and guidance overrides.
- Added the live "Will generate" interpretation line and a collapsible help table.
- Fixed: stale-offset bug in directive removal; backspace characters introduced by a patch script.

Increment 4: telemetry and history

- Added gpu_info.py (static facts from torch, live readings from nvidia-smi).
- Added db.py with a 45-column SQLite table, insert, fetch, search, get, stats and a CLI viewer.
- app.py now times each stage, samples the GPU before and after, records every image, and renders a details report.
- Added the History tab: aggregate line, search, limit, table, row click to image and details.

Increment 5: real 4K and 8K output

- Added upscaler.py: Real-ESRGAN x4plus through spandrel, half precision, 512 px tiles with 32 px overlap, multi-pass to target, Lanczos finish.
- Parser: 2k, 4k, uhd, 8k words and oversize exact resolutions set an upscale target; exact sizes are centre-cropped.
- Database: eight upscale columns added with automatic migration.
- UI: upscaled image shown in the gallery and History; details report gains an Upscale section; header shows the upscaler.
- Measured: 4K in 3.4 s, 8K in 8.5 s on the RTX 3090.
- Fixed: torchvision CPU build replaced with the CUDA build.

Documentation (22 Oct 2026)

- Added docs/ with the documentation as PDFs (one combined, one per section) and docs-src/ with the Markdown sources, figures and build script.

10.2 Roadmap

Priority	Item	Notes
High	SDXL refiner or a diffusion-based "hires fix" pass	Adds generated detail at 2x before Real-ESRGAN; better fine texture than super-resolution alone
High	Image-to-image and inpainting tabs	Same pipeline family; lets a chosen image be edited rather than regenerated
Medium	Model switcher in the prompt, e.g. "model: sdxl-turbo"	Turbo gives 1 s drafts at 4 steps; FLUX fits in 24 GB for higher quality
Medium	Export history to CSV and chart timings	pandas is already present
Medium	Thumbnail cache for the History grid	Avoids loading full PNGs when browsing
Low	Authentication and LAN access	Gradio supports basic auth; needed before exposing beyond localhost
Low	Package as a Windows service or Docker image	For always-on use
Low	Cloud deployment profile	Hugging Face Space with ZeroGPU for sharing, or a RunPod template for burst capacity